

Research Report

Proper Scoring Rules for Survival Analysis: Foundations and Later Impact

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Course: Statistical Learning - Spring 2026

Selected paper: Hiroki Yanagisawa, "Proper Scoring Rules for Survival Analysis" (ICML 2023)

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Section 1: Justification for Selected Paper

The paper I selected is “Proper Scoring Rules for Survival Analysis” by Hiroki Yanagisawa, published at ICML 2023. I selected this paper because it fits the three conditions of the course project. In my spreadsheet it was marked as not expensive to compute, reproducible with limited resources, and directly related to Statistical Machine Learning. The experiments use public survival datasets and a small neural network, so the main ideas can be tested on a normal computer. This was important for me because I wanted to reproduce a real research paper without needing a GPU cluster or private data.

The topic is also a good match for statistical learning. Survival analysis tries to predict when an event will happen, for example death, disease, customer churn, or machine failure. The difficult part is censoring. For some people or objects, the event is not observed during the study, so the exact event time is unknown. A model must learn from both complete and censored observations. This creates a statistical problem that is different from normal regression or classification.

The main contribution of Yanagisawa (2023) is to connect survival analysis with the theory of proper scoring rules. A scoring rule is a loss or evaluation measure for a predicted probability distribution. A proper rule has an important property: on average, the best prediction is the true probability distribution. The paper studies four well-known ideas - pinball loss, logarithmic score, Brier score, and ranked probability score - and extends them to right-censored survival data. It gives conditions under which these extensions are proper. The paper also shows that the logarithmic and Brier extensions work especially well in its experiments.

I think this paper is a strong choice because it has both theory and practical experiments. It is not only proposing another neural-network architecture. Instead, it asks whether the objective used to train and compare survival models is statistically meaningful. This question matters because a model can appear good under a badly chosen metric even if its predicted distribution is wrong. The paper therefore connects probability theory, statistical modeling, neural networks, calibration, and model evaluation in one clear problem.

For my implementation I reproduced the four main scoring-rule methods on the same three public datasets used by the paper: f1chain, prostateSurvival, and support. I also compared my Cen-log results with exact values reported in the paper and tested two smaller experiments about the number of time bins and the DeepHit ranking parameter. This made the paper suitable for the implementation part as well as for the literature review.

Another reason I selected this paper is that the experimental design is easy to understand and compare with the theory. The paper uses three public datasets and the same type of neural network for all of them. This makes it possible to focus on the loss functions instead of changing many parts of the model at the same time. In my reproduction, I could keep the main settings close to the paper and run the experiments on CPU. I reduced the number of epochs to make the experiments practical, but I still obtained the same general pattern: Cen-log and Cen-Brier performed better than Cen-RPS and Portnoy. This gave me a useful check that I was implementing the main idea correctly.

The paper is also suitable for the literature-review part of the project. Its main ideas are connected to earlier work on censoring, scoring rules, quantile regression, and calibration. Later papers also discuss both the strengths and the limits of the proposed scores. Because of this, I could trace a clear backward and forward research path instead of only collecting unrelated citations.

Section 2: Backward Research - The Foundations

The selected paper is part of a long research line. Its ideas come from classical survival analysis, probabilistic forecasting, censored regression, and modern machine learning. To organize this section, I used the paper's reference list as the main source and checked related works in Connected Papers and Semantic Scholar. The most important earlier works are summarized below.

2.1 Classical survival analysis and censoring

A basic starting point is the Kaplan-Meier estimator (Kaplan & Meier, 1958). It estimates a population survival curve when some observations are right-censored. It is nonparametric and is still one of the standard tools in survival analysis. Yanagisawa uses the same censoring setting and also uses a Kaplan-Meier based calibration idea. This shows that the 2023 paper is not replacing classical survival analysis. It builds a machine-learning evaluation framework on top of the same data problem.

Another major foundation is the Cox proportional hazards model (Cox, 1972). Cox introduced a flexible regression model for censored event times and made survival regression widely usable. However, the Cox model focuses on a hazard ratio and uses the proportional hazards assumption. Modern machine-learning survival models often try to predict the full event-time distribution instead. Yanagisawa focuses on this distributional view, because proper scoring rules are naturally defined for probability distributions.

2.2 Proper scoring rules as the statistical foundation

The central theoretical source for the selected paper is the general theory of proper scoring rules. Gneiting and Raftery (2007) gave an important review and development of this theory. In simple terms, a scoring rule compares a probabilistic prediction with what actually happens. A strictly proper scoring rule rewards an honest probability distribution: the expected score is best when the forecast is the true distribution. This is useful for both estimation and evaluation.

This idea becomes difficult in survival analysis because the true event time is sometimes hidden by censoring. We do not always observe the value that a normal scoring rule needs. The main question behind Yanagisawa (2023) is therefore: how can standard proper scores be changed so that they still make statistical sense when the outcome is censored?

2.3 Earlier loss functions and metrics used for survival data

Different parts of the 2023 paper come from different earlier methods. Portnoy (2003) developed censored regression quantiles. This extends quantile regression to censored outcomes and is closely connected to the pinball loss. Yanagisawa later proves properness for the Portnoy-style score under conditions related to discretization. This gives a stronger theoretical explanation for a method that was already used in censored quantile regression.

For probability prediction, the Brier score has also been important. Graf et al. (1999) developed prediction-error measures for censored survival data and helped make the IPCW Brier score and integrated Brier score standard evaluation tools. These measures became very popular because they are easy to understand: they measure squared error between predicted survival probabilities and observed outcomes. However, later work showed that common survival versions of these scores are not always proper under the desired definition.

Calibration is a second kind of evaluation, and here the main earlier work is Haider et al. (2020). They studied how to build and evaluate individual survival distributions and proposed D-calibration, which checks whether the predicted probabilities behave like a uniform distribution. Yanagisawa uses D-calibration in his experiments

and then proposes a second metric, KM-calibration, which compares the average predicted distribution with the Kaplan-Meier estimate. I used both metrics in my implementation.

Deep learning made the issue more important. DeepHit (Lee et al., 2018) directly predicts a discrete distribution of event times and uses a likelihood term together with a ranking term. Other neural survival models also introduced custom objectives. These models can be powerful, but a custom loss is not automatically a proper scoring rule. A model may improve the chosen metric without moving closer to the true event-time distribution.

2.4 The direct bridge to Yanagisawa (2023)

Rindt et al. (2022) is the closest direct foundation of the selected paper. That work studied common scoring rules for right-censored survival regression and showed that several popular measures are not proper. It also proved that the right-censored log-likelihood is proper and proposed SuMo-net, a neural model that can optimize it. This result changed the discussion from only asking “Which model has a better score?” to also asking “Is the score itself statistically valid?”

An earlier warning came from Blanche et al. (2018), who proved that the concordance index is not proper for evaluating t-year predicted risks. This was important because the concordance index had been the standard survival metric for a long time.

Yanagisawa (2023) continues exactly this question. It gives another view of the censored logarithmic score, studies assumptions caused by discretizing the event-time distribution, proves a properness result for Portnoy’s estimator, and introduces proper censored versions of the Brier score and ranked probability score. In this sense, the paper combines several older research lines into one framework.

Year	Work	Main idea connected to the selected paper
1958	Kaplan & Meier	Nonparametric survival estimation with censored data.
1972	Cox	Regression for censored event times using proportional hazards.
1999	Graf et al.	Brier-style prediction error for survival data.
2003	Portnoy	Quantile regression for censored outcomes.
2007	Gneiting & Raftery	General theory of proper and strictly proper scoring rules.
2018	Lee et al. (DeepHit)	Neural prediction of discrete event-time distributions with a ranking term.
2018	Blanche et al.	Proves that the concordance index is not proper for t-year risks.
2020	Haider et al.	D-calibration and a framework for individual survival distributions.
2022	Rindt et al.	Shows problems with common survival scores and proves properness of censored log-likelihood.
2023	Yanagisawa	Extends four major scores and gives new properness results for survival analysis.

Section 3: Forward Research - The Impact

The paper is recent, so its forward citation history is still small compared with older survival papers. However, later work shows that the topic has continued to grow. For this section, I focused on later papers that cite Yanagisawa (2023) and checked how they use its ideas, not only whether they mention it in related work. The main impact is not a single new neural architecture. It is the stronger use of proper scoring rules as a design principle for survival models and evaluation.

3.1 Later work by the same research line

A clear continuation is “Survival Analysis via Density Estimation” by Yanagisawa and Akiyama (ICML 2025). That paper changes the viewpoint again: it treats survival analysis as a density-estimation problem and then converts density outputs into survival functions. It can handle competing risks and can also discuss dependent censoring when information about the dependence is available. The paper explicitly refers back to proper scoring rules for survival analysis. In its experiments it uses the simplified censored logarithmic score from Yanagisawa (2023), and it also follows the earlier recommendation that the DeepHit ranking weight should be zero when the goal is a proper loss.

This is important because it shows that the 2023 scoring-rule work became part of a larger modeling framework. The later paper does not just repeat the same four losses. It uses proper scoring ideas as a basis for evaluating more general survival settings, including competing risks and dependence between event and censoring times.

3.2 Proper scores for competing risks

Another related direction is the work of Alberge et al. (2025) on survival analysis with competing risks. It proposes a strictly proper censoring-adjusted separable scoring rule. The loss can be optimized with stochastic methods and is used with gradient-boosting trees. The key connection to Yanagisawa is the same principle: the training objective should estimate event probabilities while remaining proper under censoring. The newer work also focuses on scalability, because a separable loss can be optimized on subsets of the data.

The movement toward competing-risk settings is a natural next step. The 2023 paper mainly studies the standard right-censored case. Real applications may have several event types, such as different causes of failure or different medical outcomes. Later work tries to keep the statistical guarantees while moving to these more complex settings.

3.3 Stricter definitions and criticism of the practical side

Not all later work accepts the 2023 method as it is, and this group taught me the most.

Sonabend et al. (2025) study properness in the external validation of survival models. They place Yanagisawa (2023) next to Rindt et al. (2022) as one of the first formal definitions of a proper survival scoring rule. Then they introduce a stricter marginal definition of properness and show that under this definition both the integrated survival Brier score and the right-censored log-likelihood are theoretically improper, although their simulations show that both still behave well in practice. Their argument is only possible because the earlier papers had already given a precise definition to work with.

Kopper et al. (2026) give the sharpest criticism I found. They accept that the score of Yanagisawa (2023) is proper, but they point out that it depends on a parameter that is not known in practice, which limits its use as a training loss. This is exactly the problem I met in my own implementation. The weights of the extended scores must be estimated from the current model with an iterative reweighting step, and the quality of training depends on how good that estimate is. In my results, Cen-log and Cen-Brier were clearly better than Cen-RPS and Portnoy, and the explanation is the same one: the first two have weights that are usually close to 0 or 1, so

an estimation error does not hurt much. Reading this criticism after finishing my experiments was the point where my own numbers and the literature agreed.

3.4 A new 2026 framework and a criticism of plug-in weights

A very direct follow-up is “Proper Scoring Rules for Right-Censored Survival Data” by Jonkers et al. (2026). This paper cites Yanagisawa (2023) and compares against its censored scores. It proposes a general construction: first apply the censoring mechanism to the predictive distribution and then use a proper score on the observed-data distribution. The authors derive censored versions of several standard scores, including pinball, Brier, log, CRPS, and energy scores.

The 2026 paper also points to an important limitation of the 2023 approach. Some Yanagisawa scores use weights that depend on unknown parts of the event-time distribution. In practice these weights may be replaced by estimates from the model being evaluated. Jonkers et al. show that forecast-dependent plug-in weighting can create ranking problems in some simulated settings. Their solution is to make the weighting depend on the censoring mechanism instead of the forecast. This is a good example of scientific impact: a later paper takes an earlier idea seriously, finds a practical weakness, and proposes a more general solution.

3.5 Influence on the broader scoring-rule literature

The 2026 Annual Review article by Waghmare and Ziegel includes survival analysis as an application area for proper scoring rules. In its discussion, it names Yanagisawa (2023) as work that proposes proper extensions of the Brier and ranked probability scores to survival data. This shows that the paper is no longer only part of a narrow survival-analysis discussion. It is also being included in broader reviews of proper scoring rules in statistics and machine learning.

The same research line is moving beyond right censoring. An ICML 2026 paper by Yanagisawa and Akiyama is titled “A Strictly Proper Scoring Rule and a Calibration Metric for Interval-Censored Data Analysis.” Interval censoring is harder because the event is known only to occur inside a time interval. The move from right-censored data to interval-censored data is a natural extension of the main question in the 2023 paper: how can we design losses and metrics that remain statistically correct when the outcome is only partly observed?

Year	Later work	How it continues the topic
2025	Yanagisawa & Akiyama	Uses proper scoring ideas in a density-estimation framework and extends to competing risks and dependent censoring.
2025	Alberge et al.	Builds a strictly proper censoring-adjusted loss for scalable survival and competing-risk learning.
2025	Sonabend et al.	Introduces a stricter marginal definition of properness and re-examines the standard survival scores.
2026	Kopper et al.	Accepts the properness result but criticises the dependence on an unknown parameter during training.
2026	Jonkers et al.	Directly compares with Yanagisawa and proposes a more general forecast-independent censoring framework.
2026	Waghmare & Ziegel	Places Yanagisawa’s survival scores inside a broad review of modern

Year	Later work	How it continues the topic
2026	Yanagisawa & Akiyama	proper scoring rules. Extends the proper-score idea to interval-censored data and calibration.

Section 4: Conclusion and Reflection

The main thing I learned from this project is that survival analysis is not only about choosing a prediction model. The loss function and the evaluation metric are also part of the statistical model. A very flexible neural network can still learn the wrong distribution if its objective rewards the wrong behavior. Proper scoring rules give a clear way to think about this problem.

The backward research showed a clear path from classical censoring methods to modern machine learning. Kaplan-Meier and Cox created basic tools for censored time-to-event data. Portnoy and Graf developed losses and evaluation ideas for censored outcomes. Gneiting and Raftery gave the general theory of proper scores. Rindt et al. then brought properness directly into neural survival regression, and Yanagisawa extended this direction to several major scoring rules.

One detail from this path surprised me. The concordance index has been a standard survival metric since 1982, but the proof that it is not proper for t-year risks appeared only in 2018. For many years, models were compared with a measure that can prefer a wrong distribution to the true one. Nobody hid this problem. The tools needed to check it simply came from a different research community.

The forward research showed that the topic is still active. Later papers are working on competing risks, dependent censoring, interval censoring, scalable optimization, and more general ways to construct proper scores. I also found that later work does not simply accept the 2023 method. Kopper et al. (2026) criticise the dependence on an unknown parameter, and Jonkers et al. (2026) study the forecast-dependent weights and propose another framework. This is useful because it shows how research develops: one paper answers a question but also creates new questions.

For a graduate student, I see both a challenge and an opportunity. The challenge is that strong ML research needs several skills at the same time: probability, statistics, programming, experiments, and clear writing. Reproducing a paper also showed me that small missing details such as preprocessing or training settings can change the final numbers. In my case my results were 0.02 to 0.08 away from the published values, and most of that gap comes from decisions the paper does not describe. The opportunity is that useful research does not always require a huge model or expensive hardware. This selected paper is a good example. Its main contribution is a careful statistical question and a theoretical analysis, supported by experiments that can be run on normal hardware.

Overall, the project made the idea of “being an academic in SML” more concrete for me. Reading only the final results of a paper can make research look clean and linear. Reproducing the work and tracing its references shows a different picture. Research is built from older ideas, practical choices, assumptions, and limitations. A useful contribution can be a new model, but it can also be a better loss, a better evaluation rule, or a clearer proof of when an existing method is valid.

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